Predicting Stellar Labels for the Sagittarius Stream Using Gaia DR3 XP Spectra

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Abstract

High-resolution spectroscopic surveys provide precise stellar parameters and chemical abundances, but cover only a fraction of the stars observed by Gaia. This project aims to transfer stellar labels from APOGEE DR17 to Gaia DR3 BP/RP XP spectra using a multi-output neural network, and then apply the trained model to candidate members of the Sagittarius stellar stream. The intended outputs are effective temperature, surface gravity, metallicity, nitrogen-to-oxygen abundance, and aluminum-to-iron abundance.

For this project stage, we implemented a runnable data-acquisition and preprocessing pipeline for the Sagittarius catalogue of Vasiliev et al. [14]. The pipeline reconstructs the 55,192-source catalogue, extracts Gaia DR2 identifiers, matches the identifiers to Gaia DR3, separates unique and ambiguous counterparts, selects sources with XP continuous spectra, and downloads XP coefficient products in restart-safe batches. The current crossmatch contains 46,014 DR2 sources with at least one DR3 candidate, including 42,386 sources with exactly one candidate and 3,628 sources with multiple candidates. Among the uniquely matched sources, 27,069 have XP continuous spectra. At the latest checkpoint, 25,850 of these XP sources had been downloaded successfully, leaving 1,219 pending because the Gaia DataLink service was temporarily returning HTTP 503 service-unavailable responses. The APOGEE DR17 and Gaia XP training sample remains the next major implementation task.

1 Introduction

The chemical and dynamical properties of stars preserve information about the assembly of the Milky Way and its accreted satellites. Metallicity and elemental-abundance ratios can distinguish stellar populations, constrain enrichment histories, and identify debris associated with disrupted dwarf galaxies. The Sagittarius dwarf spheroidal galaxy and its tidal stream provide an important setting for studying chemical evolution during an ongoing accretion event.

Detailed abundance measurements are commonly obtained from medium- and high-resolution spectroscopic surveys. APOGEE DR17 provides precise near-infrared stellar labels for a large stellar sample [1]. However, the number of likely Sagittarius stream members identified using Gaia astrometry is much larger than the number with existing spectroscopic labels. Vasiliev et al. [14] assembled a catalogue of 55,192 high-probability Sagittarius stream candidates using Gaia DR2 astrometry, 2MASS photometry, distance estimates, and available line-of-sight velocities.

Gaia DR3 provides low-resolution BP/RP spectrophotometry, commonly called XP spectra, for a much larger stellar sample [4, 3]. These spectra are represented by basis-function coefficients and contain broad spectral information across the optical and near-infrared wavelength range. Previous studies demonstrate that data-driven models can infer stellar parameters, metallicities, and selected abundance information from Gaia XP data [10, 11, 8].

The goals of this project are to:

1. construct an APOGEE DR17 and Gaia DR3 XP training sample;
2. predict $T_{\rm eff}$, $\log g$, [Fe/H], [N/O], and [Al/Fe] from normalized XP coefficients;
3. evaluate predictive accuracy and uncertainty on held-out stars; and
4. apply the validated model to Sagittarius stream candidates with XP spectra.

2 Data

2.1 Gaia DR3 XP Spectra

Gaia DR3 publishes BP and RP mean spectra using coefficients that multiply internally calibrated basis functions [3]. Following Kane et al. [8], the intended model input is constructed from 55 BP and 55 RP coefficients. Each coefficient group will be divided by its first coefficient, after which the two unit-valued first coefficients will be removed. This gives

$$N_{\rm XP} = (55 - 1) + (55 - 1) = 108 \tag{1}$$

input features per source.

The current pipeline retrieves XP continuous coefficient products through the Gaia DataLink service. Downloads are divided into small batches, saved immediately as FITS files, retried after temporary errors, and recursively divided into smaller requests after repeated failures. Existing FITS files are scanned before each run so completed sources are skipped when the notebook resumes.

2.2 APOGEE DR17 Labels

The labeled training sample will be drawn from APOGEE DR17 [1]. The intended target vector is

$$\mathbf{y}_i = (T_{\rm eff}, \log g, [\mathrm{Fe/H}], [\mathrm{N/O}], [\mathrm{Al/Fe}])_i \, . \tag{2}$$

The APOGEE DR17 and Gaia DR3 XP overlap has not yet been implemented in the submitted code. The next stage will load the APOGEE DR17 `allStar` catalogue, identify Gaia counterparts, retrieve corresponding XP coefficients, and apply documented quality cuts. Planned checks include ASPCAP quality flags, signal-to-noise ratio, finite labels, and selected effective-temperature and surface-gravity ranges. The exact APOGEE field used for [N/O] will be verified against the APOGEE DR17 data model before training.

2.3 Sagittarius Stream Catalogue

The application sample comes from the Sagittarius stream catalogue of Vasiliev et al. [14]. Because this catalogue uses Gaia DR2 source identifiers, the identifiers cannot be treated as interchangeable with Gaia DR3 identifiers. The pipeline uploads the DR2 identifiers to the Gaia Archive and joins them against `gaiadr3.dr2_neighbourhood` and `gaiadr3.gaia_source`.

The original VizieR transfer was interrupted near the end of the table. A smaller tail request recovered the missing records. The two pieces were combined by record number, duplicate records were removed, and the result was checked against the complete sequence from 1 through 55,192. The reconstructed catalogue contained all 55,192 expected rows and zero missing record numbers. A local FITS copy is used on subsequent runs so the pipeline does not depend on repeated VizieR access.

3 Implemented Solution

The runnable notebook `src/reading_catalog_ps4.ipynb` performs the following tasks:

1. load the reconstructed Vasiliev et al. catalogue;
2. extract numeric Gaia DR2 identifiers from `SimbadName`;
3. upload these identifiers and query the Gaia DR2-to-DR3 neighborhood table;
4. count candidate DR3 counterparts for each DR2 source;
5. separate exactly-one-candidate and multiple-candidate matches;
6. retain unique DR3 matches with `has_xp_continuous=True`;
7. download XP continuous coefficient products in restart-safe batches;
8. save failed and missing source lists; and
9. combine downloaded FITS batches and remove duplicate source rows.

The downloader was updated for the current Gaia DataLink interface by using `DATAMODEL_GAIA` rather than the deprecated `RAW` data structure. Intermediate products are saved under `data/`, including all crossmatch candidates, unique matches, ambiguous matches, the XP source list, and summary files. Large generated FITS products can be reproduced by rerunning the notebook and therefore do not need to be stored in normal Git history.

4 Preliminary Results

Table 1 summarizes the current catalogue, crossmatch, and XP download results.

Table 1: Current Sagittarius catalogue and Gaia DR3 XP pipeline results.

Measurement	Count
Sagittarius catalogue sources	55,192
DR2 sources with at least one DR3 candidate	46,014
DR2 sources with exactly one DR3 candidate	42,386
DR2 sources with multiple DR3 candidates	3,628
DR2 sources without a returned DR3 candidate	9,178
Unique DR3 matches with XP continuous spectra	27,069
XP availability among unique matches	63.86%
XP sources downloaded at latest checkpoint	25,850
XP sources pending at latest checkpoint	1,219
Download completeness	95.50%

The current match counts differ from an earlier preliminary result of 50,074 unambiguous matches. The present Gaia query returned 46,014 DR2 sources with at least one candidate, of which 42,386 had exactly one candidate. This difference will be investigated before the final analysis by checking query behavior, Gaia Archive release changes, source-identifier coverage, and the interpretation of unmatched versus ambiguous sources.

The ambiguous-match histograms show a concentrated set of candidates near zero angular separation and near zero magnitude difference, together with broader distributions at larger separations and magnitude differences. The current pipeline conservatively retains only sources with exactly one DR3 candidate. This decision reduces the risk of attaching XP coefficients to an incorrect counterpart, although it may exclude some valid sources with multiple candidates.

At the latest checkpoint, 25,850 of 27,069 selected XP sources had been retrieved. This is 95.50% of the target sample. The remaining 1,219 requests were delayed by temporary HTTP 503 responses from the Gaia DataLink service. Because every completed batch is saved immediately and existing source identifiers are skipped, the remaining download can resume without repeating completed work.

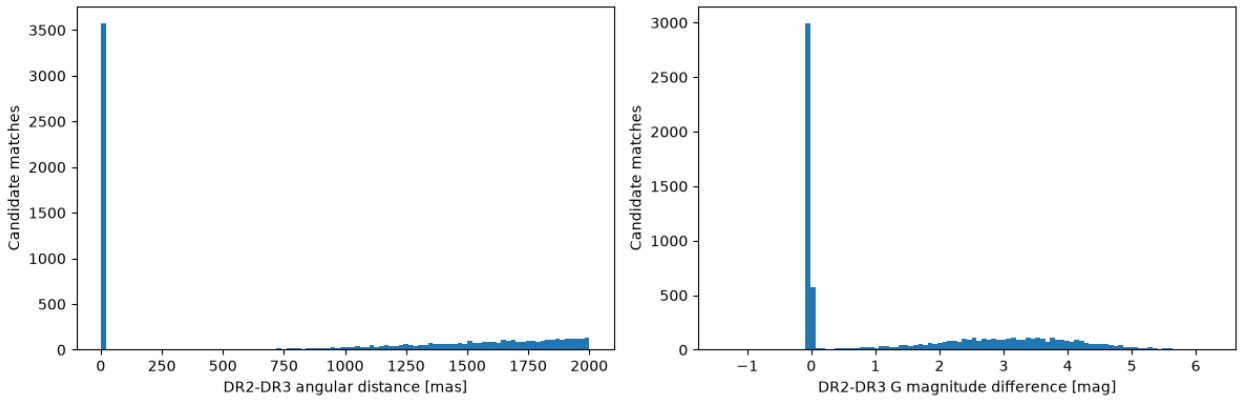


Figure 1: Angular-distance and magnitude-difference distributions for ambiguous Gaia DR2-to-DR3 candidate matches.

5 Planned Neural-Network Solution

The proposed model is a multi-output multilayer perceptron based on Kane et al. [8]. The initial model will receive 108 normalized XP coefficients and use two fully connected hidden layers with 128 units each and dropout probability $p = 0.2$. A baseline implementation will first predict one value per target using mean squared error. A refined model will predict a mean and variance for each target and minimize a heteroscedastic Gaussian negative log-likelihood:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{5} \left[\frac{(y_{ij} - \mu_{ij})^2}{2\sigma_{ij}^2} + \frac{1}{2} \log(\sigma_{ij}^2) \right]. \tag{3}$$

Dropout will also support approximate epistemic-uncertainty estimation through repeated stochastic forward passes [5]. The distinction between aleatoric and epistemic uncertainty follows Kendall and Gal [9]. The implementation will use PyTorch [12], with NumPy, SciPy, scikit-learn, Astropy, and Matplotlib supporting preprocessing, splitting, evaluation, astronomy-specific file handling, and visualization [6, 15, 13, 2, 7].

The APOGEE and Gaia overlap will be split reproducibly into training, validation, and test subsets. Feature and target transformations will be fitted only on the training subset to avoid information leakage. Evaluation will include mean absolute error, root mean squared error, predicted-versus-true plots, residual distributions, and uncertainty calibration for each label.

6 Limitations and Next Steps

The current implementation establishes the Sagittarius application dataset but does not yet provide a trained preliminary model. The next tasks are:

1. finish the remaining 1,219 XP downloads and regenerate the final summary;
2. create `data/data.txt` with source links and reproduction instructions;
3. construct the APOGEE DR17 and Gaia XP overlap sample;
4. document all ASPCAP, signal-to-noise, temperature, gravity, and finite-label cuts;
5. report the number of stars retained after each cut;
6. plot the distributions of all five target labels;
7. create reproducible training, validation, and test splits;
8. train a baseline regression model and report held-out metrics; and
9. apply the validated model to the Sagittarius XP sample.

7 Conclusion

This stage produced a runnable, documented, and restart-safe data pipeline for the Sagittarius stream application sample. The pipeline reconstructed all 55,192 catalogue records, extracted Gaia DR2 identifiers, performed a Gaia DR2-to-DR3 crossmatch, identified 42,386 unique candidate matches, and selected 27,069 sources with XP continuous spectra. At the latest checkpoint, 25,850 XP sources had been downloaded successfully, corresponding to 95.50% of the selected sample. The remaining downloads can resume after the Gaia service becomes available. The primary next step is construction of the APOGEE DR17 and Gaia XP training sample, followed by preprocessing, model training, and held-out evaluation.

Code and Data Availability

The runnable implementation and data-reproduction instructions are available in the project repository at https://github.com/LightsRedemption/DeepLearning_TeamAstro.

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